Ali Ahmadi Esfidi

AI Researcher & Developer

aliahmadiesfidi@outlook.com | Mr-Ahmadi | Ali Ahmadi Esfidi | Portfolio

Education

B.Sc. Computer Science › Amirkabir University of Technology — Sep 2022 – Sep 2026

- Location: Tehran, Iran • Field of study: Computer Science • GPA: 18.93/20 • Top 20%

High School Diploma › Dr. Shahriari High School (SAMPAD) — Sep 2019 – Jun 2022

- Location: Qom, Iran • Field of study: Mathematics & Physics • GPA: 18.42/20

Research Interests

- Deep Reinforcement Learning
- Machine Learning for Decision-Making Systems
- AI for Infrastructure Monitoring
- Optimization Methods for AI and Networking
- Computational Biology & Bioinformatics
- Software-Defined Networks & Resource Allocation

Experiences

Research Assistant › Supervisor: Prof. Ghatee — Jan 2025 – Present

NORC Lab, Amirkabir University of Technology

- **Led** an AI team and **engineered** web systems and deep learning models for automated bridge damage detection in collaboration with the Tehran Urban Planning and Research Center.
- **Gained hands-on experience** in developing efficient deep-learning pipelines for computer vision applications, focusing on model compactness and structural health monitoring.
- *Stack:* `PyTorch` • `OpenCV` • `ReactJS` • `NodeJS` • `PostgreSQL` • `Docker`

Research Assistant › Supervisor: Prof. Khonsari — Sep 2024 – Present

High Performance Networks Lab, Tehran University

- **Developed** deep reinforcement learning systems for complex agricultural decision-making, focusing on reward engineering and policy optimization to balance multiple objectives.
- **Acquired proficiency** in benchmarking reinforcement learning algorithms across discrete and continuous action spaces, and in utilizing simulation environments for training and evaluation.
- *Stack:* `PyTorch` • `TensorFlow` • `Pandas` • `NumPy`

Research Assistant › Supervisor: Dr. Dolati — Sep 2025 – Present

Sharif University of Technology

- **Formulated** MDPs for switch pairing and collaborative rule caching, and designed simulation environments for joint scheduling-routing policies in programmable data planes.
- **Implemented** optimization frameworks for in-network aggregation in distributed machine learning, focusing on ILP decomposition, flexible routing strategies, and scalable algorithmic design.
- *Stack:* `PyTorch` • `Numba` • `Pandas` • `NumPy`

Teaching Assistant › Graduate & Undergraduate Courses — Sep 2024 – Present

Amirkabir University of Technology & Tehran University

Data Mining B.Sc.	Dr. Yousefimehr	Sep 2026 – Present
Quantum Information Processing M.Sc.	Prof. Khonsari	Sep 2025 – Jan 2026
Computational Data Mining M.Sc.	Prof. Ghatee	Sep 2025 – Jan 2026
Introduction to Logic B.Sc.	Dr. Didehvar	Sep 2025 – Jan 2026
Theory of Computation B.Sc.	Dr. Didehvar	Sep 2024 – Jun 2026
Artificial Intelligence & Workshop B.Sc.	Prof. Ghatee, Dr. Yousefimehr	Sep 2024 – Jan 2025

Publications

- M. Saberi, A. Ahmadi Esfidi, M. Dolati, N. Omidvar, **FlexINA: In-Network Aggregation for Accelerating Distributed Machine Learning with Flexible Routing**, Published in *IEEE Transactions on Network and Service Management*. `View Paper`.

- P. Heidari, A. Khonsari, A. Ahmadi Esfidi, A. Mehrvarz, E. Khodaei, A. Dadlani, **DeepIrrigo: Deep Reinforcement Learning for Continuous and Discrete Action Irrigation Optimization — A Case Study in Iran**, Under review at *Intelligent Systems with Applications*. View Paper.
- M. Saberi, A. Ahmadi Esfidi, M. Dolati, A. Khonsari, A. Movaghar, **Rule Caching in Programmable Networks with Deep Reinforcement Learning**, Submitted to *IEEE Transactions on Network and Service Management*.
- P. Heidari, A. Ahmadi Esfidi, A. Mehrvarz, E. Khodaei, A. Khonsari, **Irrigation Optimization in Agricultural Fields Using Deep Reinforcement Learning Approaches**, Published in *CSICC 2025*. View Paper.

Selected Projects

See more

DRL-Based Beamforming Optimization for Movable Intelligent Surfaces

Jun 2026 – Present

- **Designed** a DRL framework to optimize beamforming for MIS-assisted SWIPT systems, jointly configuring phase shifts and power splitting ratios to maximize energy harvesting.
- *Stack:* PyTorch • Gymnasium • MOSEK • NumPy

RL-Based ML Job Scheduler with Distribution Optimization

Aug 2025 – Oct 2025

- **Developed** a two-stage hierarchical PPO system for scheduling ML jobs on multi-server accelerator clusters using the Gavel dataset, reducing average job completion time.
- *Stack:* PyTorch • Gymnasium • NumPy

Machine Learning Perspectives on Gold-Binding Peptides

May 2025 – Jun 2025

- **Applied** machine learning to classify and predict gold-binding affinity of peptide sequences, comparing regression and classification approaches across multiple featurization strategies.
- *Stack:* PyTorch • Scikit-learn • Hugging Face • Pandas • NumPy/SciPy

RNA Pairing Pattern Recognition Model

Mar 2024 – Jul 2024

- **Developed** a system to predict RNA secondary structures using SCFGs, evolutionary information, and enhanced CYK parsing for complex motifs; presented at CBRC Journal Club.
- *Stack:* NetworkX • Biopython • PhyML • NumPy/SciPy

Skills

Programming Languages	Python • JavaScript • TypeScript • R • Java • C++ • Swift
Frameworks	React • Node.js • Express.js • Django • Electron
Data Science / ML	PyTorch • TensorFlow • Pandas • Scikit-learn • NumPy • OpenCV • LangChain
Platforms & Tools	Docker • Git • GitHub • MongoDB • PostgreSQL • Hugging Face

Certifications

Bioinformatics Internship Program › Biocan — Nov 2025
Scientific Secretary: Dr. Z. Salehi • Course Director: Dr. K. Kavousi

Introduction to Bioinformatics › Biocan — Jul 2025
Scientific Chair: Dr. K. Kavousi • Score: 96/100 • Top 3% of 350+ students

Scrum Foundations Course › Ultima Training Tech — Dec 2024
Instructor: Josef Balahan

Volunteering

Student Scientific Magazine "Halghe" › Amirkabir University of Technology — Dec 2025 – Present
Contributed to scientific editing and supported the editorial team

Technical & Web Support Volunteer › Amirkabir University of Technology — May 2023, Feb 2025
Web and procurement support for DAI DAY (2025) and the International Conference on AI and Smart Vehicles (2023)